

CCS516 Assignment 2 (Chan Khai Shen's Part)

1 FUZZY LOGIC SYSTEM FOR PREDICTING LIKELIHOOD OF CUSTOMER TO BUY A PRODUCT

The fuzzy logic system is used for predicting the likelihood of a customer to buy a product based on the product price, product review and delivery time. The fuzzy logic system is implemented without using any library and tool box. The method for implementing fuzzification, rule inference and defuzzification in the code is explained in this section.

1.1 FUZZIFICATION

Fuzzification is implemented by applying different mathematical formula for crisp input in different ranges. The code uses if-else to separate the crisp input based on the range of the value and apply different mathematical formulas. The formula for product price low, product price normal, product price high, product review bad, product review average, product review good, delivery time slow and delivery time fast are expressed below.

$$\text{product price low} = \begin{cases} 1, & \text{product price} \leq 20 \\ -0.1 \times \text{product price} + 3, & 20 < \text{product price} \leq 30 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{product price normal} = \begin{cases} 0, & \text{product price} \leq 20 \text{ or } \text{product price} \geq 70 \\ 0.1 \times \text{product price} - 2, & 20 < \text{product price} \leq 30 \\ 1, & 30 < \text{product price} \leq 60 \\ -0.1 \times \text{product price} + 7, & \text{otherwise} \end{cases}$$

$$\text{product price high} = \begin{cases} 0, & \text{product price} \leq 60 \\ 0.1 \times \text{product price} - 6, & 60 < \text{product price} \leq 70 \\ 1, & \text{otherwise} \end{cases}$$

$$\text{product review bad} = \begin{cases} 1, & \text{product review} \leq 1.5 \\ -\frac{2}{3} \times \text{product review} + 2, & 1.5 < \text{product review} \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{product review average} = \begin{cases} 0, & \text{product review} \leq 1.5 \text{ or } \text{product review} \geq 4 \\ \text{product review} - 1.5, & 1.5 < \text{product review} \leq 2.5 \\ 1, & 2.5 < \text{product review} \leq 3 \\ -1 \times \text{product review} + 4, & \text{otherwise} \end{cases}$$

$$\text{product review good} = \begin{cases} 0, & \text{product review} \leq 3 \\ \text{product review} - 3, & 3 < \text{product review} \leq 4 \\ 1, & \text{otherwise} \end{cases}$$

$$\text{delivery time slow} = \begin{cases} 1, & \text{delivery time} \leq 3 \\ -0.5 \times \text{product review} + 2.5, & 3 < \text{delivery time} \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

$$delivery\ time\ fast = \begin{cases} 0, & delivery\ time \leq 3 \\ 0.5 \times delivery\ time - 1.5, & 3 < delivery\ time \leq 5 \\ 1, & otherwise \end{cases}$$

1.2 RULE INFERENCE

Rule inference is implemented based on Mamdani inference method. The formula for calculating the rule strength of each rule is expressed below.

$$Rule\ 1\ strength = Product\ price\ low$$

$$Rule\ 2\ strength = \min (product\ review\ good, delivery\ time\ fast)$$

$$Rule\ 3\ strength = \min (product\ review\ bad, 1 - product\ price\ low)$$

$$Rule\ 4\ strength$$

$$= \min (product\ price\ normal, product\ review\ good, delivery\ time\ slow)$$

$$Rule\ 5\ strength = \min (product\ price\ normal, product\ review\ average)$$

$$Rule\ 6\ strength$$

$$= \min (product\ price\ high, product\ review\ good, delivery\ time\ slow)$$

$$Rule\ 7\ strength$$

$$= \min (product\ price\ high, product\ review\ average, delivery\ time\ fast)$$

The degree of each class (less likely, normal, more likely) of the fuzzy output (likeliness of customer to buy product) are determined based on the formulae below.

$$More\ likely = \max (rule\ 1\ strength, rule\ 2\ strength)$$

$$Less\ likely = Rule\ 3\ strength$$

$$Normal = \max (rule\ 4\ strength, rule\ 5\ strength, rule\ 6\ strength, rule\ 7\ strength)$$

1.3 DEFUZZIFICATION

The fuzzy value of the likeliness of customer to buy product is de-fuzzified into crisp value using centroid of area function. Below is the formula used to calculate the centroid of area.

Centroid of gravity (crisp value of likeliness)

$$\begin{aligned} & (0 + 1 + 2) \times lesslikely + 3 \times \min(0.5, \max(less\ likely, normal)) + (4 + 5 + 6) \times normal + \\ & 7 \times \min(0.5, \max(normal, more\ likely)) + (8 + 9 + 10) \times more\ likely \\ = & \frac{3 \times less\ likely + \min(0.5, \max(less\ likely, normal)) + 3 \times normal + \min(0.5, \max(normal, more\ likely)) + 3 \times more\ likely}{} \end{aligned}$$